1 Lesson Overview

In this lesson, students will build a professional Power BI dashboard using Spain Electricity Demand time series data. The dashboard focuses on KPI development, time-series analysis, hourly demand patterns, and monthly demand trends.

1.1 Learning Outcomes

By the end of this lesson, students will be able to:

- Create a professional Power BI dashboard from scratch.
- Build KPI cards using DAX measures.
- Analyze electricity demand trends using time series data.
- Develop hourly and monthly demand visualizations.
- Apply dashboard design best practices.
- Generate business insights from electricity demand data.

2 Project Information

2.1 Dataset Overview

The project uses Spain Electricity Demand data collected over multiple years. The dataset contains actual electricity demand measurements and time-related variables that allow us to perform time series analysis.

2.2 Important Fields

Field	Purpose
time	Time-series analysis
total load actual	Electricity demand value
hour	Hourly analysis
month	Monthly analysis
year	Filtering and comparison

Table 1: Important Fields Used in the Dashboard

3 Dashboard Canvas Design

3.1 Canvas Settings

Navigate to:

Format Page → Canvas Settings

Property	Value
Width	1500 px
Height	820 px
Background Alignment	Middle
Transparency	85%

Table 2: Canvas Configuration

3.2 Design Objective

The objective is to create a clean executive-level dashboard that provides a quick overview of electricity demand patterns across Spain.

4 Creating the Dashboard Header

4.1 Insert Header Shape

Insert a Rounded Rectangle shape at the top of the dashboard.

Property	Value
Width	1405 px
Height	65 px
Shape Type	Rounded Rectangle
Fill Color	#573B92

Table 3: Header Shape Settings

4.2 Dashboard Title

Insert a text box containing:

```
SPAIN ELECTRICITY DEMAND DASHBOARD
Page 1 - Electricity Demand Overview
```

4.3 Title Formatting

Property	Value
Font Size	20
Font Color	White
Bold	Enabled
Effects	Off
Alignment	Center

Table 4: Dashboard Title Formatting

5 Creating KPI Measures

5.1 Maximum Demand Measure

Navigate to:
Modeling → New Measure

```
Max Demand =
MAX('cleaned_energy_weather_for_powerbi'[total load actual])
```

5.2 Minimum Demand Measure

```
Min Demand =
MIN('cleaned_energy_weather_for_powerbi'[total load actual])
```

5.3 Average Demand Measure

```
Avg Demand =
AVERAGE('cleaned_energy_weather_for_powerbi'[total load actual])
```

6 Designing KPI Cards

6.1 Required KPI Cards

Create three card visuals:

1. Maximum Demand
2. Minimum Demand
3. Average Demand

6.2 Card Formatting

Property	Value
Shape	Rounded Rectangle
Corner Radius	10%
Display Units	None

Table 5: Card Shape Settings

6.3 Callout Value Formatting

Property	Value
Font Size	30
Bold	Yes
Alignment	Center
Color	Dark Purple

Table 6: Callout Value Settings

6.4 Label Formatting

Property	Value
Font	Segoe UI Semibold
Font Size	11
Position	Below Value
Color	Brown Tone
Horizontal Space	4

Table 7: Label Settings

7 Actual Electricity Demand Over Time

7.1 Visual Container Design

Insert a Rounded Rectangle shape behind the chart.

Property	Value
Width	565 px
Height	310 px
Fill Color	White
Transparency	6%
Border Width	3 px
Border Color	#573B92
Glow Effect	Enabled

Table 8: Container Design

7.2 Create Line Chart

Field Area	Value
X-axis	Time
Y-axis	Average Total Load Actual

Table 9: Chart Fields

7.3 Formatting

Setting	Value
Background	Off
X-axis Title	Off
Y-axis Title	Off
Gridlines	Off
Line Color	#EB895F
Highest Point Color	#73361C

Table 10: Line Chart Formatting

8 Hourly Average Demand Analysis

8.1 Chart Creation

Create an Area Chart or Line Chart.

8.2 Field Configuration

Field Area	Value
X-axis	Hour
Y-axis	Average Total Load Actual

Table 11: Hourly Demand Configuration

8.3 Chart Formatting

Property	Value
Background	Off
Gridlines	Off
Line Color	#EB895F
Area Transparency	25%

Table 12: Hourly Demand Formatting

8.4 Business Interpretation

Students should observe:

- Lower demand during early morning hours.
- Increasing demand during business hours.
- Peak demand during evening periods.
- Daily consumption cycle patterns.

9 Monthly Average Demand Analysis

9.1 Chart Creation

Create a Clustered Column Chart.

9.2 Field Configuration

Field Area	Value
X-axis	Month
Y-axis	Average Total Load Actual

Table 13: Monthly Demand Configuration

9.3 Color Formatting

Element	Color
Regular Bars	#EB895F
Highest Month	#73361C

Table 14: Monthly Demand Colors

10 Interactive Filtering

10.1 Year Slicer

Create a slicer using the Year field to allow users to filter demand across different years.

10.2 Month Slicer

Create a slicer using the Month field to allow monthly filtering.

10.3 Slicer Formatting

Property	Value
Style	Tile
Selected Color	#573B92
Unselected Color	Light Gray
Corner Radius	10

Table 15: Slicer Formatting

11 Key Insight Panel

11.1 Creating the Insight Area

Add a rectangle shape below the charts to communicate business findings.

11.2 Insight Title

Key Insight

11.3 Business Insight

Electricity demand demonstrates strong hourly and monthly seasonality. Consumption increases significantly during working hours and evening periods, while monthly fluctuations suggest seasonal demand behavior across Spain.

12 Dashboard KPI Summary

KPI	Business Purpose
Maximum Demand	Identify highest electricity usage
Minimum Demand	Identify lowest electricity usage
Average Demand	Understand overall consumption
Hourly Average Demand	Analyze daily demand patterns
Monthly Average Demand	Analyze seasonal demand patterns
Demand Over Time	Evaluate long-term trends

Table 16: Dashboard KPI Summary

13 Lesson Summary

In this lesson, students developed a professional Spain Electricity Demand Dashboard using Power BI. They created KPI measures, designed executive-level KPI cards, analyzed demand behavior over time, explored hourly and monthly trends, applied modern dashboard design principles, and derived meaningful business insights from time series data.